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Promoting Early Relational Health in Primary Care: A Scoping Review of Interventions and Evidence Gaps

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Abstract

Background: Early relational health – the quality of caregiver–child interactions – is foundational for socio-emotional development and long-term wellbeing. The American Academy of Pediatrics has called for a paradigm shift toward relationship-centered care in pediatric primary care, yet a gap remains between this vision and real-world implementation.

Objectives: To map early relational health interventions implemented in pediatric primary care, summarize reported outcomes and identify implementation strategies, barriers, and facilitators.

Data Sources: PubMed, CINAHL, and PsycINFO were searched for studies published between 2014 and 2024, supplemented by backward citation searching.

Study Eligibility Criteria, Participants, and Interventions: A scoping review methodology following PRISMA-ScR guidelines was used. Data were charted and synthesized using descriptive and thematic analysis.

Results: Eighteen studies met inclusion criteria. Interventions were delivered in clinic, home, or hybrid settings and were commonly integrated into routine workflows. Reported outcomes included improvements in caregiver responsiveness, parenting self-efficacy, and child language

and socio-emotional development. Implementation strategies included staff training, workflow integration, fidelity monitoring, and community partnerships. Key facilitators were contextual responsiveness, strong relational supports, and alignment with clinical processes. Barriers included small or homogeneous samples, reliance on parent-report outcomes, short follow-up periods, and resource constraints.

Conclusions and Implications of Key Findings: Early relational health interventions are feasible and show promise in pediatric primary care when embedded into routine care and adapted to context. Addressing methodological and implementation challenges is essential to advancing relationship-centered pediatric practice.

Scoping Review Registration: Open Science Framework (OSF) DOI:

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What's New

This scoping review advances pediatric primary care by mapping early relational health interventions, highlighting feasible, workflow-integrated approaches that improve caregiver and child outcomes, and identifying key implementation strategies, barriers, and evidence gaps to guide scalable relationship-centered care.

Promoting Early Relational Health in Primary Care: A Scoping Review of Interventions and Evidence Gaps

The earliest years of life lay the foundation for lifelong health and well-being, with early caregiver–child interactions shaping brain architecture, stress regulation, and socio-emotional development.^{1,2} Responsive, nurturing relationships promote secure attachment, self-regulation, and resilience, while stress, social disadvantage, and limited access to supports can challenge caregivers' capacity to provide these interactions.^{3,4,5} Recognizing the centrality of relationships

for healthy development, the American Academy of Pediatrics (AAP) has called for a paradigm shift from a deficit-focused model centered on adverse childhood experiences toward a strengths-based approach that elevates safe, stable, and nurturing relationships as a biological necessity.⁶

Early relational health (ERH) has emerged as a unifying framework for understanding and strengthening caregiver–child interactions across health and community settings.⁶ ERH emphasizes moment-to-moment behaviors such as attuned communication, mutual engagement, and responsive caregiving that support cognitive, socio-emotional, and physical health.^{7,8} Activities such as shared play, reading, holding, and feeding build the foundations for emotional security and positive developmental trajectories.⁹ This ecobiodevelopmental perspective highlights how early experiences become biologically embedded and how relational supports buffer the effects of toxic stress.⁶

Pediatric primary care is uniquely positioned to advance this relational health vision. With near-universal access and frequent visits in the early years, primary care offers timely, low-stigma opportunities to promote responsive caregiving through anticipatory guidance, modeling, and brief relational interventions.^{10,11} Pediatric clinicians, including pediatricians, family physicians, nurse practitioners, and physician assistants, are trusted professionals who consistently engage with families from birth through early childhood.¹² Programs such as PlayReadVIP (formerly Video Interaction Project) and Family Check-Up illustrate the potential for embedding relationally focused strategies into routine visits, demonstrating improvements in caregiver responsiveness, interaction quality, and child outcomes.^{13,14} The AAP further advocates for vertical integration (primary, secondary, tertiary prevention) and horizontal integration (cross-sector collaboration) to scale relational health promotion across systems.

Despite growing interest and the emergence of promising individual models, research on relationally focused interventions in pediatric primary care remains fragmented and lacks comprehensive synthesis. Although numerous parenting and behavioral programs have been evaluated in these settings, few explicitly frame their approaches within the ERH paradigm, examine how relational processes are operationalized and strengthened in practice, or incorporate standardized approaches to assessing relational health to guide intervention selection.^{6,51}

Existing reviews illustrate this gap. A recent scoping review cataloged a broad range of parenting programs delivered in primary care but paid limited attention to critical implementation factors – such as clinic workflows, provider–caregiver relationships, and team structures – that influence adoption and fidelity.¹¹ Without examining these contextual and implementation elements, even effective interventions risk remaining siloed, inconsistently applied, or unsustainable. Similarly, meta-analyses and other reviews^{15,16,17,18,19} have demonstrated benefits of preventive interventions in primary care for parenting and developmental outcomes but have not synthesized how interventions explicitly target early relational processes, measure relational health outcomes, or integrate into pediatric workflows. Moreover, relational health research spans multiple disciplines, including infant mental health, home visitation, and pediatrics, contributing to a fragmented evidence base.

Together, these gaps underscore the need for a new comprehensive review that moves beyond cataloging existing interventions to systematically analyze how ERH principles are embedded in pediatric systems, what strategies support scalability and which barriers impede long-term success. In addition, limited attention has been given to how relational health or parenting assessment tools may support pediatric providers in identifying family needs and selecting interventions matched to levels of relational risk or support.⁵¹ To advance the AAP's

relational health vision and inform scalable approaches, it is essential to understand how ERH interventions are designed, implemented, and sustained within real-world pediatric systems. A scoping review is therefore warranted to map the landscape of ERH interventions, characterize their implementation strategies, and identify contextual factors that shape feasibility, sustainability, and impact.

Purpose of this Study/Objective

Given the central role of pediatric primary care in promoting ERH, there is a need to better understand how ERH interventions are implemented, delivered, and sustained within routine care. This scoping review synthesizes evidence on interventions designed to strengthen caregiver-child relationships in pediatric primary care settings, including approaches focused on responsive caregiving, shared emotional connection, relational guidance, coaching, and structured supports delivered during pediatric encounters.

Aligned with AAP guidance advocating for the integration of developmental and relational promotion into routine pediatric care,²⁰ this review focuses on understanding what is being implemented, how it is delivered, and under what conditions it succeeds. Specifically, the review addresses the following research questions:

1. What early relational health interventions have been implemented in pediatric primary care?
2. What are the characteristics, reported outcomes, and implementation strategies?
3. What barriers and facilitators to implementation have been identified?

By mapping the existing literature, this review clarifies how ERH principles are currently embedded in pediatric practices, identifies common implementation challenges and supports, and

highlights opportunities to strengthen the integration of relationship-centered care into pediatric health systems.

Methods

This scoping review systematically identified and synthesized evidence on ERH interventions in pediatric primary care. Scoping reviews are appropriate for examining the breadth of literature, identifying research gaps, and informing future directions.²¹⁻²³ The review follows the methodological framework outlined by Arksey and O'Malley,²¹ refined by Levac et al.,²⁴ and further developed by the Joanna Briggs Institute (JBI).²³

The protocol was registered on the Open Science Framework (OSF),²⁵ and reporting adhered to the PRISMA Extension for Scoping Reviews (PRISMA-ScR) to ensure transparency and completeness.²⁶

Search Strategy and Study Selection

A comprehensive search strategy was developed in consultation with a health science librarian to enhance rigor and reproducibility.²⁸ Searches were conducted in October 2024 across PubMed, CINAHL, and PsycINFO for literature published between 2014 and 2024. Search terms included combinations of keywords and Medical Subject Headings (MeSH) related to children, caregivers, early relational health, interventions, and pediatric primary care (see Table 1).²⁹

Database searches identified 1,116 records across PubMed, CINAHL, and PsycINFO. After duplicate removal, 922 records remained for title and abstract screening. Following exclusion based on predefined eligibility criteria, 62 articles underwent additional review, and 17 full-text articles were assessed for eligibility. One additional study was identified through backward citation searing, resulting in 18 studies included in the final review (see PRISMA flow diagram, Figure 1).

Screening was conducted using Covidence (Veritas Health Innovation, n.d.) and followed PRISMA-ScR guidelines.^{26,27} A two-step screening process was used. First, the first author and independent reviewers screened titles and abstracts based on predefined inclusion criteria: (1) evaluation of an ERH intervention for children aged 0-5 years; (2) measurement of child outcomes (e.g., socio-emotional development, language, attachment); (3) implementation within pediatric primary care settings. Second, the first and third authors independently reviewed full-text articles identified as potentially eligible following title and abstract screening. Discrepancies were resolved through discussion and consensus. Eighteen studies met inclusion criteria and were included in the final review.

Inclusion and Exclusion Criteria

Studies were included if they described interventions promoting ERH (e.g., parent-child connection, attunement, co-regulation) for children ages 0–5, were implemented or initiated in pediatric primary care settings (e.g., well-child visits, integrated behavioral health clinics, family medicine), and reported intervention components, outcomes, or implementation factors. Eligible studies were published in English between 2014 and 2024 and included peer-reviewed empirical research or relevant gray literature. Studies were excluded if they focused solely on physical health without relational components, targeted children outside the 0-5 age range, or were conducted outside primary care settings (e.g., specialty care, inpatient units, home-based programs, schools, community centers, or telehealth-only models). Systematic and scoping reviews were also excluded.

Data Extraction and Coding

Data were extracted using a structured charting form aligned with research questions.^{21,24} Extracted variables included study characteristics, intervention features, target populations,

outcomes, implementation-related information, and whether studies reported approaches to assessing relational health or parenting behaviors.

Data analysis combined deductive and inductive approaches.^{30,31} Deductive coding applied *a priori* categories informed by the research questions and relevant frameworks, including the Expert Recommendations for Implementing Change (ERIC) taxonomy,³² to categorize intervention components, outcomes, and implementation strategies. Inductive thematic analysis captured contextualized barriers and facilitators, including factors related to workflow alignment, provider engagement, and family responsiveness. Three reviewers independently conducted open coding, with discrepancies resolved through consensus.³³

Interrater Reliability of Barrier and Facilitator Coding

Three coders independently applied the finalized codebook to 29 barrier excerpts and 20 facilitator excerpts. For barriers, mean agreement was 0.92 (SD = 0.17), with Krippendorff's alpha of approximately 0.90. For facilitators, mean agreement was 0.93 (SD = 0.14), with Krippendorff's alpha of approximately 0.91, indicating strong reliability across domains. Coding discrepancies were resolved through discussion prior to thematic synthesis.

Results

Overview of Included Studies

Study Identification and Selection

Eighteen studies from fifteen journals were included (see Table 2). Most were conducted in the United States, with additional studies from Africa, the United Kingdom, Australia, and Iceland. All examined ERH interventions delivered in pediatric primary care settings or integrated clinical-community models. Target populations were parent-child dyads with infants or young children. Interventions ranged from book distribution programs (e.g., Reach Out and

Read) and video-based parent coaching (e.g., Video Interaction Project) to multi-component home visiting and internet-supported approaches, reflecting diverse strategies to promote caregiver engagement and early development in clinical or linked community contexts.

Study Design and Populations

Most studies used randomized controlled trial designs,³⁴⁻³⁷ with others employing qualitative methods,³⁸ pre-post designs,^{39,40} and cross-sectional analyses.⁴¹ Sample sizes ranged from small pilot samples (n=19 families)⁴⁴ to large multi-site trials exceeding 1,000 participants.⁴² Samples primarily consisted of mother-infant or mother-child dyads from birth to preschool age. Several studies focused on underserved populations, including low-income Latino and African American families,^{43,44} though others were skewed toward middle-to-high socioeconomic status families³⁴ or lacked detailed demographic reporting. Few studies explicitly included fathers or other caregivers.

Intervention Characteristics, Outcomes, and Implementation Strategies

Intervention Characteristics

Interventions were most often delivered in pediatric clinics^{35,38,42-46} or community health centers,^{40,47} with some extending beyond the clinic through home-based components^{34,37} or hybrid clinic-home models.^{36,48} Implementing personnel included pediatricians,^{38,43-46} nurses,^{40,47,50} child development specialists,^{36,45} psychologists,⁴⁹ nurse practitioners,³⁹ pediatric residents,³⁸ trainees,³⁴ and community health workers.⁴⁸ Delivery modes included in-person visits, home visits, group sessions, phone follow-up, mailed materials, and multimedia supports such as video feedback or online peer support groups.^{34,45,47,49} Intervention intensity ranged from single-session models embedded within well-child visit^{38,46} to longitudinal programs sustained across multiple years.^{34,35,39,44,45,49} Theoretical foundations varied, with some interventions

explicitly grounded in Social Learning, Attachment Theory, Social Cognitive, or Family Systems theories,^{34,40,42} while others did not specify a guiding framework. Literacy promotion programs commonly drew on the Reach Out and Read model.^{38,43-45}

Reported Outcomes

Across studies, interventions demonstrated positive caregiver-, child-, and system-level outcomes. Caregiver outcomes included improved parenting knowledge, confidence, and responsive behaviors,^{35,45} as well as increased engagement in shared reading, language-rich interactions, and play.^{36,43,45} Child outcomes most frequently included gains in language, socio-emotional development, and school readiness,^{35,45} assessed using standardized tools such as the Preschool Language Scale, MacArthur-Bates Communicative Development Inventories, Bracken School Readiness Assessment, Ages & Stages Questionnaires: Social-Emotional (ASQ:SE), and the Child Behavior Checklist (CBCL).^{35,45} Several studies also reported reductions in depressive symptoms.^{36,43,45} At the system level, interventions were associated with increased anticipatory guidance, improved care coordination, and higher preventive service utilization.^{36,43,45} Where assessed, implementation outcomes indicated high feasibility, acceptability, and satisfaction among caregivers and providers (see Table 3).^{15,35}

Implementation Strategies

Implementation strategies varied widely and spanned training, delivery, and sustainment. Common approaches included staff training and supervision, though intensity ranged from brief workshops to ongoing models.^{34,39,40,48,49} Fidelity monitoring varied, with some studies using structured observation or audit-and-feedback systems and others relying on self-report.^{34,36,37,47} Many interventions were embedded into pediatric workflows, leveraging well-child visits or screening processes,^{38,42,43,50} while others emphasized contextual through bilingual materials or

community adaptations.^{38,48} Tools such as books, handouts, coaching scripts, and digital supports reinforced intervention content beyond clinic visits.^{35,41,45-47} System-level strategies included workflow integration, task-sharing, and calls for reimbursement and quality improvement to enhance sustainability.^{36,37,43}

Thematic Analysis of Barriers and Facilitators

Across studies, implementation success was shaped by methodological, contextual, and relational factors. Barriers included small or homogeneous samples limiting generalizability,^{34,35} exclusion of fathers or eligibility restrictions,^{34,38,48} reliance on parent-reported outcomes without blinding,^{36,41,43,49} short follow-up periods,^{48,49} and limited use of control groups.^{40,41,48} Practical constraints such as time pressures, competing demands, limited provider buy-in, and weak fidelity monitoring impeded implementation.^{37,42,43,46} Parent-level challenges (e.g., low engagement, language barriers) and structural disruptions (e.g., funding limitations, COVID-19, technology access) also affected delivery and evaluations.^{37,43,47}

Facilitators included culturally and contextually responsive delivery, such as bilingual materials, home-based or hybrid models, and low-cost formats that improved engagement.^{34,35,45,48} Strong relational supports – characterized by trust, clinician modeling, coaching, and peer support – were central to caregiver participation and behavior change.^{42,43,46,48-50} Embedding interventions within routine pediatric care or established community infrastructure, supported by training, supervision, and fidelity monitoring, enhanced feasibility, scalability, and sustainability across settings.^{34,36-39,41,44,45,48,49}

Discussion

Overview and Impact of the Work

By framing pediatric parenting and behavioral interventions within an ERH lens, this scoping review clarifies what is being implemented in pediatric primary care, how interventions are delivered, and under what conditions they succeed. This perspective strengthens translation of the AAP's relational health paradigm into practical, relationship-centered strategies integrated into routine well-child care. To our knowledge, this is the first review to synthesize pediatric interventions explicitly through an ERH lens, bridging pediatric care, developmental-behavioral strategies, and community-based supports that are often examined separately.

Findings demonstrate that ERH-focused interventions are feasible and promising in pediatric primary care. Across studies, brief, relationship-centered supports embedded into routine well-child visits strengthened caregiver-child interactions, enhanced parenting self-efficacy, and improved child socio-emotional and cognitive outcomes.^{34,42,49} Interventions aligned with existing workflows – such as anticipatory guidance, modeling, or in-visit coaching – were particularly scalable and sustainable.

The review also highlights several key impacts. First, ERH interventions demonstrate real-world relevance by addressing common family challenges – such as stress, limited resources, and inconsistent access to services – through trusted pediatric settings. Second, they yield two-generation benefits, with caregiver well-being improving alongside child development.^{6,49} Third, findings operationalize AAP guidance by demonstrating how nurturing, responsive relationships can be strengthened within routine care when interventions are contextually responsive and workflow-supported. At the same time, promoting relational health may also require identifying and addressing maladaptive or harmful parenting behaviors, including harsh or punitive discipline strategies, to ensure families receive developmentally appropriate supports matched to their relational needs.⁵¹⁻⁵³

Together, these findings position ERH as a unifying framework for pediatric primary care, clarifying implementation conditions and offering a roadmap for scaling relationally grounded approaches within healthcare systems.

Paradigm Shift and Provider Engagement

Despite promising evidence, implementation requires a shift from a primarily medicalized model toward developmental wellness and relationship-centered care. This transition, endorsed by the AAP, necessitates integration of relational health into clinical priorities, quality metrics, and reimbursement structures. Provider buy-in is essential; clinicians must view relational health promotion as preventative care rather than an optional add-on. Although strengths-based approaches are foundational to ERH, pediatric primary care may also benefit from approaches that identify maladaptive parenting behaviors alongside relational assets to guide intervention planning and support family needs more comprehensively.⁵¹⁻⁵³ Evidence from this review reinforces that ERH interventions are linked to improvements in language, socio-emotional development, and school readiness, underscoring their relevance to long-term health trajectories.

Feasibility and Promise of ERH Interventions

ERH interventions span a flexible continuum of intensity, from brief, low-dose approaches to ongoing, multicomponent models. Brief interventions, such as *Reach Out and Read (ROR)* and *PlayReadVIP* (formerly Video Interaction Project (VIP)) integrate directly into well-child visits, require minimal resources, demonstrate high reach and feasibility.^{34,36,43} Despite their brevity, these models positively influence caregiver behaviors and interaction environments.

In contrast, more intensive approaches, including *Parent Connex*, and *Parent-Child Interaction Therapy (PCIT)* adaptations, provide sustained coaching and more structured behavioral health support, with the potential for more durable gains but requiring greater coordination, clinical capacity, and investment.^{42,49} These models were not represented among the studies meeting eligibility criteria for this review, highlighting an evidence gap within this synthesis rather than a lack of established effectiveness in the broader literature.

Similarly, other widely implemented pediatric primary care models such as *HealthySteps* were not included in the final set of eligible studies despite their strong presence in practice-based systems and their alignment with relationship-centered care. Together, these omissions underscore that the present review reflects a subset of ERH-relevant interventions meeting specific inclusion criteria rather than a comprehensive accounting of established models.

Across this continuum, emerging integrated care approaches further illustrate how relational and behavioral supports can be embedded within pediatric systems. Advanced primary care models and multilevel behavioral health integration frameworks demonstrate the feasibility of combining universal relational promotion with targeted assessment and intervention pathways within routine care delivery.^{51,53-55}

A tiered or stepped-care approach may help reconcile these tradeoffs. In this context, universal relational health promotion strategies can be complemented by more targeted supports when parenting or relational concerns are identified through developmental surveillance, screening, or caregiver report.^{52,53} This structure allows brief, high-reach interventions to function as the foundation while reserving higher-intensity services for families with greater needs. Such models have demonstrated effectiveness and efficiency in community and school-based systems by improving reach while optimizing allocation of resources. Applying similar

strategies within pediatric primary care may better align relational health promotion with population health goals while supporting sustainability of implementation.

Implementation Challenges and Variability

Implementation varied widely across delivery models, workforce composition, and contextual supports. Interventions were led by clinicians, community health workers (CHWs), or co-located specialists with variability in fidelity monitoring, training, and workflow integration.^{34,42,45}

Common barriers included attrition, limited resources, lack of blinding, COVID-19 disruptions, and inconsistent follow-up.^{40,49} Beyond logistics, cultural barriers – particularly misalignment with clinic priorities – often limited integration. Without leadership and institutional alignment, interventions risk remaining peripheral rather than embedded in routine care. Integrating access to standardized parenting or relational health assessments into pediatric workflows may further support implementation by improving consistency in identifying family needs and informing referral pathways, though feasibility and time demands remain important considerations in busy primary care settings.⁵²

Facilitators included culturally and linguistically responsive materials, integration into routine visits, strong caregiver–provider relationships, clinician modeling, and peer support.^{34,43} Context also shaped feasibility: CHW-led or hybrid models may be better suited to rural or low-resource settings, whereas co-located behavioral health aligns more readily with urban practices.^{37,42} Sustainable implementation further requires navigating tensions between mission and reimbursement, reinforcing the need for policy and payment structures that support relational care.^{6,12}

Key Elements for Effective ERH Integration in Pediatric Practice

Several elements supported successful implementation: trust and relationship-building,^{34,49} embedding interventions into existing care structures,^{36,43} cultural and linguistic inclusivity,^{38,40} and practical supports for staff – including structured training, ongoing coaching, and manageable time demands.^{42,48} Effective relationship-centered care includes both supporting family strengths and identifying harmful or maladaptive interaction patterns that may interfere with healthy caregiver-child relationships.⁵¹ Conversely, limited follow-up, small samples, insufficient provider engagement, and poor contextual fit constrained impact.^{44,49} Importantly, no single ERH model emerged as universally “best”. Effectiveness depended on alignment between intervention design and practice context. Advancing relationship-centered care therefore requires flexible adaptation, leadership, provider engagement, and structural supports that enable clinics to select approaches suited to local needs.

Training and Workforce Development

Sustained ERH integration depends on workforce capacity. Clinicians require skills in relationship-centered communication and coaching deliverable within routine visits. Evidence indicates that structured training, ongoing coaching, and fidelity monitoring enhance clinician confidence and adherence.^{42,48} Integrating ERH principles into medical education, residency training, and continuing professional development can normalize relational health as a core component of pediatric care rather than an optional add-on, facilitating the broader paradigm shift advocated by the AAP. In addition to intervention delivery skills, providers may also benefit from familiarity with relational health or parenting assessment tools that can support care planning when parenting or relational concerns emerge.⁵²⁻⁵⁴

Implications for Research and Policy

Future research should establish consistent operational definitions of ERH outcomes,^{3,6} prioritize implementation and cost-effectiveness studies and examine mechanisms linking relational supports to child development outcomes.^{34,43}

Greater attention is needed to understand how relational health and parenting assessment tools can be used pragmatically in pediatric primary care when concerns about parenting stress, discipline practices, or caregiver-child interactions emerge. Existing evidence suggests that assessment approaches measuring both positive and maladaptive parenting behaviors may support more clinical decision-making in situations where relational or parenting concerns are identified.⁵¹⁻⁵⁴

Making these tools accessible within pediatric systems may support more targeted identification of relational concerns, improve alignment between family needs and intervention strategies, and strengthen implementation of relationship-centered care across diverse practice settings.^{52,53} Additional research is needed to determine the feasibility, acceptability, and long-term impact of incorporating relational health assessments into routine pediatric practice given workflow and time constraints. Policy efforts should embed relational health metrics into pediatric quality frameworks, incentivize team-based care, and support cross-sector partnerships to address family needs.^{36,40} These steps align with AAP's vision of pediatric care as a platform for relational health promotion and population-level well-being.

Strengths and Limitations

This review synthesizes a diverse set of parenting and behavioral interventions through the lens of ERH, examining both intervention characteristics and implementation strategies across pediatric primary care settings. Strengths include the systematic focus on relational outcomes, attention to feasibility and scalability, and integration of barriers and facilitators to

inform real-world practice. Limitations include variability in study designs, sample sizes, follow-up periods, and outcome measures, as well as potential publication and language biases. Differences in delivery agents, fidelity, and contextual supports across studies limit direct comparisons, underscoring the need for more standardized reporting and rigorous implementation research. Future research should address the lack of consistent operational definitions of ERH and develop pragmatic measures that can be feasibly integrated into clinical workflows.

Conclusion

This review demonstrates that ERH interventions in pediatric primary care are both feasible and promising. Brief, relationship-centered supports and ongoing multicomponent models can strengthen caregiver–child interactions and developmental outcomes, particularly when implemented with trust, integration into workflows, cultural and linguistic responsiveness, and practical staff supports. Realizing the potential of ERH requires aligning relational and equity-focused goals with sustainable funding and infrastructure, ensuring interventions are both effective and scalable across diverse pediatric settings. Future efforts should prioritize pragmatic measurement strategies, provider engagement, and policy mechanisms that embed relational health into pediatric quality frameworks, advancing AAP’s vision of pediatric care as a platform for developmental wellness and toxic stress prevention.

References

1. National Scientific Council on the Developing Child. The Timing and Quality of Early Experiences Combine to Shape Brain Architecture. Working Paper No. 5. Center on the Developing Child, Harvard University; 2007. Accessed August, 2025.
2. Shonkoff JP, Garner AS. The lifelong effects of early childhood adversity and toxic stress. *Pediatrics*. 2012;129(1):e232-e246. doi:10.1542/peds.2011-2663
3. Shonkoff JP, Richter L, van der Gaag J, Bhutta ZA. An integrated scientific framework for child survival and early childhood development. *Pediatrics*. 2012;129(2):e460-e472. doi:10.1542/peds.2011-0366
4. Evans GW, Kim P. Childhood poverty, chronic stress, self-regulation, and coping. *Child Dev Perspect*. 2013;7(1):43-48. doi:10.1111/cdep.12013
5. Garg A, Toy S, Tripodis Y, Silverstein M, Freeman E. Addressing social determinants of health at well-child care visits: a cluster randomized trial. *Pediatrics*. 2015;135(2):e296-e304. doi:10.1542/peds.2014-2888
6. Garner AS, Yogman MW. Preventing childhood toxic stress: partnering with families and communities to promote relational health. *Pediatrics*. 2021;148(2):e2021052582. doi:10.1542/peds.2021-052582
7. Willis E, Eddy S. Early Relational Health for Primary Care Providers. Institute for Child Success; 2022.
8. Cassidy J, Jones JD, Shaver PR. Contributions of attachment theory and research: a framework for future research, translation, and policy. *Dev Psychopathol*. 2013;25(4 Pt 2):1415-1434. doi:10.1017/S0954579413000692
9. Zauche LH, Thul TA, Mahoney AED, Stapel-Wax JL. Influence of language nutrition on children's language and cognitive development: an integrated review. *Early Child Res Q*. 2016;36:318-333.
10. Doyle O, Harmon CP, Heckman JJ, Tremblay RE. Investing in early human development: timing and economic efficiency. *Econ Hum Biol*. 2009;7(1):1-6. doi:10.1016/j.ehb.2009.01.002
11. Marie-Mitchell A, Delgado C, Gilgoff R. Parenting education to improve relational health through pediatric primary care: a scoping review. *J Prim Care Community Health*. 2024;15:21501319241306302. doi:10.1177/21501319241306302
12. Council on Early Childhood. Addressing early childhood emotional and behavioral problems. *Pediatrics*. 2016;138(6):e20163025. doi:10.1542/peds.2016-3025

13. Mendelsohn AL, Huberman HS, Berkule SB, Brockmeyer CA, Morrow LM, Dreyer BP. Primary care strategies for promoting parent-child interactions and school readiness in at-risk families. *Arch Pediatr Adolesc Med*. 2011;165(1):33-41.
doi:10.1001/archpediatrics.2010.254
14. Shaw DS, Sitnick S, Brennan LM, et al. Long-term effectiveness of the Family Check-Up on school-age conduct problems. *Dev Psychopathol*. 2016;28(4 Pt 2):1471-1486.
doi:10.1017/S0954579415001212
15. Shah R, Kennedy S, Clark MD, Bauer SC, Schwartz A. Primary care-based interventions to promote positive parenting behaviors: a meta-analysis. *Pediatrics*. 2016;137(5):e20153393.
doi:10.1542/peds.2015-3393
16. Asarnow JR, Rozenman M, Wiblin J, Zeltzer L. Integrated medical-behavioral care compared with usual primary care. *JAMA Pediatr*. 2015;169(10):929-937.
doi:10.1001/jamapediatrics.2015.1141
17. Brown CH, Wang W, Kellam SG, et al. Methods for testing theory and evaluating impact in randomized field trials. *Drug Alcohol Depend*. 2008;95(suppl 1):S74-S104.
doi:10.1016/j.drugalcdep.2007.11.013
18. Brino KAS. Pediatric mental health and the power of primary care. *J Pediatr Health Care*. 2020;34(2):E12-E20.
19. Glascoe FP, Trimm F. Brief approaches to developmental-behavioral promotion. *Pediatrics*. 2014;133(2):884-897. doi:10.1542/peds.2013-1859
20. Weitzman C, Wegner L. Promoting optimal development. *Pediatrics*. 2015;135(2):384-395.
doi:10.1542/peds.2014-3716
21. Arksey H, O'Malley L. Scoping studies: towards a methodological framework. *Int J Soc Res Methodol*. 2005;8(1):19-32.
22. Grant MJ, Booth A. A typology of reviews. *Health Info Libr J*. 2009;26(2):91-108.
23. Peters MDJ, Godfrey C, McInerney P, et al. Scoping reviews. In: *JBIManual for Evidence Synthesis*. JBI; 2020.
24. Levac D, Colquhoun H, O'Brien KK. Scoping studies: advancing the methodology. *Implement Sci*. 2010;5:69.
25. Center for Open Science. Open Science Framework. Published 2013.
26. Tricco AC, Lillie E, Zarin W, et al. PRISMA extension for scoping reviews. *Ann Intern Med*. 2018;169(7):467-473.

27. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews. *PLoS Med*. 2009;6(7):e1000097. doi:10.1371/journal.pmed.1000097
28. McGowan J, Sampson M, Salzwedel DM, et al. PRESS peer review of electronic search strategies. *J Clin Epidemiol*. 2016;75:40-46.
29. Bramer WM, Giustini D, de Jonge GB, Holland L. Better search strategies. *Res Synth Methods*. 2018;9(4):453-458.
30. Bradley EH, Curry LA, Devers KJ. Qualitative data analysis. *Health Serv Res*. 2007;42(4):1758-1772.
31. Braun V, Clarke V. Using thematic analysis in psychology. *Qual Res Psychol*. 2006;3(2):77-101.
32. Powell BJ, Waltz TJ, Chinman MJ, et al. A refined compilation of implementation strategies. *Implement Sci*. 2015;10:21.
33. Miles MB, Huberman AM, Saldaña J. *Qualitative Data Analysis*. 3rd ed. SAGE Publications; 2014.
34. Blizzard AM, Barroso NE, Ramos FG, Graziano PA, Bagner DM. Behavioral parent training in infancy. *J Clin Child Adolesc Psychol*. 2018;47(sup1):S341-S353.
35. Canfield CF, Weisleder A, Cates CB, et al. Primary care parenting intervention. *J Dev Behav Pediatr*. 2015;36(8):586-593.
36. Mendelsohn AL, Dreyer BP, Brockmeyer CA, et al. Randomized controlled trial of pediatric parenting programs. *Arch Pediatr Adolesc Med*. 2011;165(1):42-48.
37. Rockers PC, Leppänen JM, Tarullo A, et al. Community health worker intervention. *PLoS Med*. 2023;20(4):e1004222.
38. Byington C, Hobson W, Olson L, et al. The Good Habit of Reading. *J Health Care Poor Underserved*. 2008;19(2):363-368.
39. McMenamy J, Sheldrick RC, Perrin EC. Early intervention for disruptive behavior. *J Pediatr Health Care*. 2011;25(2):77-86.
40. Skuladottir A, Sigurdardottir AO, Svavarsdottir EK. Better sleep better well-being. *Nordic J Nurs Res*. 2021;42(2):85-92.
41. Rikin S, Glatt K, Simpson P, et al. Reading frequency in Reach Out and Read. *Acad Pediatr*. 2015;15(6):651-657.

42. Eismann EA, Vaughn LM, Vilvens HL, Sauls S, Darden PM. Parent perspectives on coaching. *J Child Fam Stud*. 2021;30(7):1965-1978.
43. Jimenez ME, Uthirasamy N, Hemler JR, et al. Maximizing Reach Out and Read. *Pediatr Res*. 2024;95(7):1644-1648.
44. Needlman R, Toker KH, Dreyer BP, Klass P, Mendelsohn AL. Effectiveness of reading aloud. *Ambul Pediatr*. 2005;5(4):209-215.
45. Mendelsohn AL, Dreyer BP, Flynn V, et al. Videotaped interactions in pediatric care. *J Dev Behav Pediatr*. 2005;26(1):34-41.
46. Scholer SJ, Hudnut-Beumler J, Dietrich MS. Physician–parent discussions on discipline. *Clin Pediatr*. 2011;50(8):712-719.
47. Sawyer MG, Reece CE, Bowering K, et al. Internet-based support for new mothers. *J Med Internet Res*. 2017;19(7):e258.
48. Luoto JE, Lopez Garcia I, Abubakar A, et al. Group-based parenting interventions. *Lancet Glob Health*. 2021;9(3):e309-e319.
49. Fonagy P, Sleded M, Baradon T. Parent–infant psychotherapy. *Infant Ment Health J*. 2016;37(2):97-114.
50. Hallas D, Koslap-Petraco M, Fletcher J. Social-emotional development of toddlers. *J Pediatr Nurs*. 2017;33:33-40.
51. Sausen, K.A., Randolph, J.W., Casciato, A.N., Dietrich, M.S., Scholer, S.J. The development, preliminary validation, and clinical application of the Quick Parenting Assessment, *Prev Sci*. (2021);22(6):761-771. Doi:0.1007/s11121-021-01222-5.
52. Wee S.E., Dietrich M.S., Adkins A. et al. Missed opportunities to support parents' use of healthy discipline. *Child Abuse Negl*. 2025;158:107089. Doi:10.1016/j.chiabu.2024.107089.
53. Cooke A., Attar T., Carr V. et al. Integrating a parenting assessment into practice: Pediatric providers' time and perspectives. *Matern Child Health J*. 2024;28(9): 1448-1456. Doi:10.1007/s10995-024-03885-4.
54. McCord M, Fierman A, Sisco S, et al. A two-generation, early childhood advanced primary care model. *Pediatrics*. 2024;153(4):e2023062567. doi:10.1542/peds.2023-062567.
55. Miller EB, Roby E, Zhang Y, et al. Promoting cognitive stimulation in parents across infancy and toddlerhood: a randomized clinical trial. *J Pediatr*. 2023;255:159-165.e4. doi:10.1016/j.jpeds.2022.11.013.

Table 1: Search Strategy Terms

Concept	Search Terms (TIAB)	MeSH Terms
Population: Young Children	"infant" OR "toddler" OR "preschool"	"child, preschool"
Population: Caregiver/Parents	"foster parents" OR "guardians" OR "primary caregivers"	"parents" OR "caregivers"
Population: Healthcare Providers	"nurse practitioners" OR "nurses" OR "primary care providers"	"pediatricians" OR "family physicians" OR "healthcare providers"
Early Relational Health/Parent-Child Relationships	"early relational health" OR "caregiver-child relationship" OR "attachment" OR "social-emotional development" OR "secure attachment"	"parent-child relations"
Parent-Focused Interventions	"parent support" OR "parent-child intervention" OR "responsive parenting" OR "attachment-based intervention" OR "relationship-based intervention" OR "parent-child interaction therapy" OR "dyadic intervention" OR "trauma-informed care" OR "developmental intervention" OR "social-emotional intervention" OR "parent* coaching" OR "parent* education"	"developmental screening"
Primary Care/Healthcare Context	"primary care" OR "pediatric primary care" OR "pediatric well-child visit" OR "pediatric practice" OR "telehealth" OR "home visits" OR "home-based care" OR "community clinics"	"community health services" OR "telemedicine" OR "outpatient clinics"
Access/Implementation	"insurance coverage" OR "low-resource settings" OR "high-resource settings" OR "implementation barriers" OR "healthcare delivery" OR "healthcare guidelines"	"healthcare access" OR "health disparities"

Table 2: Summary of Reviewed Articles

Citation	Country	Design	Sample	Intervention	Implementation Strategies	Facilitators	Barriers
Blizzard et al., 2018	United States	Randomized controlled trial	60 mother-infant dyads (infants	Infant Behavior Program (IBP)	Parent training; home visits; daily practice; compensation; screening;	Bilingual delivery; home-based care	Homogeneous sample; low father

			aged 12-15 months)		therapist training; supervision; randomization; fidelity checks; standardized measures; primary care recruitment		engagement
Byington et al., 2008	United States	Qualitative	Mother-child dyads (children aged 6 months-5 years)	Reach Out and Read (ROR)	Literacy promotion in well-child visits; culturally tailored communication; Spanish/bilingual materials; clinic children's library; bilingual librarian; monthly story time/events; thank-you note tracking	Language accessibility ; bilingual staff/materials; consistent access to resources/services	Language barrier
Canfield et al., 2015	United States	Randomized controlled trial	675 mother-infant dyads (infants aged birth-24 months)	Video Interaction Project (VIP) and Building Blocks (BB)	Pediatric primary care integration; trained interventionists; take-home materials; mailed intervention; standardized protocol	Parent-interventionist relationship; cultural sensitivity	Limited generalizability; no long-term follow-up; small sample; retention issues
Eismann et al., 2021	United States	Mixed Methods	11 clinics (1301 families coached, 387 families surveyed, 27 families interviewed)	Parent Connex	Provider training; routine screening; Motivational Interview-based referral; co-located parenting specialists; parent coaching; resource navigation; primary care integration	Parent-interventionist relationship; individualized support	Self-report; time constraints ; no control group; racial disparities

Fonagy, Sleed, & Baradon, 2016	United Kingdom	Randomized controlled trial	76 Parents with infant < 12 months	Psychoanalytic Parent-Infant Psychotherapy (PIP)	Referral via health/social care; eligibility screening; randomization; home visits; manualized therapy (PIP); trained therapists; weekly–biweekly sessions; therapist supervision; blinded raters; 6–12 mo follow-up; flexible dosing; standard care access	Parent–interventionist relationship; integration with standard care; consistent access to services	Recruitment challenges; unblinded staff; language barriers; self-report; limited generalizability; retention issues; no long-term follow-up
Hallas, Koslap-Petraco, & Fletcher, 2017	United States	Randomized controlled trial	5 pediatric primary healthcare offices (64 mother-toddler dyads)	Pediatric primary care office-based videotaped (DVD) parenting skills intervention	Staff training; clinic recruitment; flyers; informed consent process; randomization/blinding; secure coded data; DVD-based education; waiting room intervention; primary care integration; IRB/confidentiality procedures	Brief intervention; parent–interventionist relationship	Low engagement; retention issues; limited generalizability
Jimenez et al., 2023	United States	Baseline analysis from randomized controlled trial	440 parent-child dyads (children aged 6-9 months)	Reach Out and Read (ROR)	Clinician modeling during visits; Reach Out and Read components (books, guidance, modeling); literacy promotion in well-child care; training/Quality Improvement emphasis; validated	Clinician modeling; scalability	COVID; self-report; limited generalizability; confounding factors

					measures; system supports (visit structure, reimbursement)		
Luoto et al., 2021	Africa	Randomized controlled trial	60 villages (1200 mother-child (6-24 months) dyads)	Early Childhood Development (ECD) Programmes	Task-shifting; train-the-trainer; pre-service + ongoing training; structured curriculum; materials; supervision/fidelity checks; incentives; community engagement; tailored delivery; father engagement efforts; cultural adaptation	Buy-in; Community Health Volunteer infrastructure; peer support	Low father engagement; no long-term follow-up; self-report
McMenamy, Sheldrick, & Perrin, 2011	United States	Pre-Post (feasibility and effectiveness study)	Two primary care clinics (23 families, children aged 2-3 years)	Parenting Resource and Education Project (PREP)	Ongoing screening; identify champions; staff recruitment/training; educational materials; stakeholder training; incentives; service site changes; clinical data sharing	Buy-in; positive response	Language barrier; small sample; self-report; limited generalizability
Mendelsohn et al., 2005	United States	Randomized controlled trial	93 mother-newborn dyads	Video Interaction Project (VIP)	Service site changes; educational materials; audit & feedback; staff training; specialist deployment; clinical data sharing; low-cost scalable model	Bilingual delivery; cultural sensitivity; low-cost design	No long-term follow-up; time constraints
Mendelsohn et al., 2011	United States	Randomized	410 mother-	Video Interaction Project	Service site changes; audit & feedback (VIP);	Low-cost design	No long-term follow-up;

		controlled trial	newborn dyads	(VIP) and Building Blocks (BB)	parent self-monitoring/planning (BB); learning materials; specialist training/deployment; clinical data sharing; task-sharing		self-report; time constraints; limited generalizability
Mendelsohn et al., 2011	United States	Randomized controlled trial	675 mother-newborn dyads	Video Interaction Project (VIP) and Building Blocks (BB)	Service site changes; audit & feedback; parent self-monitoring; educational materials; specialist training/deployment; workflow integration	Bilingual delivery; cultural sensitivity	No long-term follow-up; limited generalizability
Needlman et al., 2005	United States	Pre-post intervention study	19 pediatric primary care clinics (1647 families)	Reach Out and Read (ROR)	Educational materials; stakeholder training; service site changes; reminder systems; clinical data sharing; ongoing assessment/feedback	Low cost; large study size	Selection bias
Rikin et al., 2015	United States	Cross-sectional study	6 clinics (236 caregivers, children aged 6-59 months)	Reach Out and Read (ROR)	Educational materials; outreach visits; clinician modeling/mentoring; reminder systems; family coaching; tailored intervention	Pediatrician provision of books; strong association with shared reading	No control group; self-report; confounding variables; selection bias
Rockers et al., 2023	Africa	Randomized controlled trial	764 mother-child dyads (children aged birth-24 months)	Home visit intervention integrated into community health	Service site changes; educational materials; stakeholder training; audit & feedback; incentives/supports; task-sharing;	Community Health Volunteer infrastructure	COVID

				worker operations	tailored strategies		
Sawyer et al., 2017	Australia	Randomized controlled trial	6 family health clinics (819 mother-infant dyads)	Nurse-moderated internet-based group intervention + clinic-based postnatal health check	Service site changes; educational materials; stakeholder training; digital tools; audit & feedback; clinical data sharing; incentives/supports	Peer support; parent–interventionist relationship	Homogeneous sample; no cost-effectiveness analysis
Scholer, Hudnut-Beumler, & Dietrich, 2011	United States	Randomized controlled trial	258 caregiver-child dyad (children aged 12 months-5 years)	Play Nicely Program	Educational strategies; interpersonal communication; workflow integration; tailoring; scalability supports	Positive response; parent–interventionist relationship	Time constraints; personnel not blinded
Skuladottir, Olafía Sigurdardóttir, & Kolburn Svavarsdóttir, 2022	Iceland	Pre-post intervention study	114 caregiver-infant dyads	Better Sleep Better Well-Being (FES-BSBW)	Advanced practice nurse training/use; face-to-face education; telephone follow-up; evidence-based design; nurse-led routine care integration	Parent–interventionist relationship; positive response	Language barrier; small sample; low father engagement; no long-term follow-up; limited generalizability

Table 3: Reported Outcomes

Articles	Intervention	Key Findings
Blizzard et al., 2018	Infant Behavior Program (IBP)	Improved responsive caregiving and parenting “do” skills across all time points; reduced intrusiveness and “don’t” parenting behaviors at post-intervention and 3-month follow-up
Byington et al., 2008	Reach Out and Read (ROR)	Parents reported gratitude for book provision, perceived benefits for child development and family interaction, and positive perceptions of clinic experience and staff
Canfield et al., 2015	Video Interaction Project (VIP) and Building Blocks (BB)	VIP reduced physical punishment at 24 months, mediated by increased responsive parenting and reduced psychosocial risk; no significant effects for BB
Eismann et al., 2021	Parent Connex	Significant improvements in parenting role satisfaction, parenting efficacy, and reduced parenting stress; high satisfaction with tailored and nonjudgmental support
Fonagy, Slead, & Baradon, 2016	Psychoanalytic Parent-Infant Psychotherapy (PIP)	No significant differences in infant development or attachment; improvements in maternal mental health, reduced parenting stress, and more positive parent-infant relationship representations
Hallas, Koslap-Petraco, & Fletcher, 2017	Pediatric primary care office-based videotaped (DVD) parenting skills intervention	Both groups improved social-emotional outcomes; intervention group showed greater gains in maternal confidence, though between-group differences were not statistically significant
Jimenez et al., 2023	Reach Out and Read (ROR)	Dose-response relationship observed; exposure to multiple components associated with stronger home literacy environments and improved clinician modeling of literacy behaviors
Luoto et al., 2021	Early Childhood Development (ECD) Programmes	Improvements in child cognition, language, and socioemotional development; group-only delivery improved cognition and socioemotional outcomes, with evidence of peer effects and dose-response associations
McMenamy, Sheldrick, & Perrin, 2011	Parenting Resource and Education Project (PREP)	High caregiver satisfaction; reductions in parenting stress and harsh/inconsistent discipline; improvements in positive parenting sustained at 6-month follow-up
Mendelsohn et al., 2005	Video Interaction Project (VIP)	Improved cognitive and expressive language outcomes, particularly for children of mothers with 7-11 years of education; reduced need for early intervention (non-significant trend)
Mendelsohn et al., 2011	Video Interaction Project (VIP) and Building Blocks (BB)	VIP improved parent-child interaction quality and home reading behaviors, especially among higher literacy

		mothers; BB showed moderate improvements; effects attenuated among lowest literacy group
Mendelsohn et al., 2011	Video Interaction Project (VIP) and Building Blocks (BB)	VIP reduced infant daily media exposure and delayed initial exposure; effects mediated by increased cognitive among higher-literacy families
Needlman et al., 2005	Reach Out and Read (ROR)	Increased literacy-promoting behaviors, particularly among lower-education families and Black and Latino caregivers; most associations remained significant in adjusted models
Rikin et al., 2015	Reach Out and Read (ROR)	Receipt of > 4 books associated with increased reading frequency; predictors included bedtime routines, home book availability, caregiver interest, and higher education
Rockers et al., 2023	Home visit intervention integrated into community health worker operations	No overall effects on growth or developmental outcomes; improvements observed visual processing, EEG measures, and caregiver engagement; adherent participants showed gains in growth and motor development
Sawyer et al., 2017	Nurse-moderated internet-based group intervention + clinic-based postnatal health check	Generally non-interior outcomes between groups; slight reductions in language scores and transient increases in parental isolation in internet-supported group
Scholer, Hudnut-Beumler, & Dietrich, 2011	Play Nicely Program	Significant increases in caregiver intention to use non-physical discipline; strongest effects observed with combined multimedia and clinician counseling exposure
Skuladottir, Olafia Sigurdardottir, & Kolburn Svavarsdottir, 2022	Better Sleep Better Well-Being (FES-BSBW)	Improved maternal confidence, family support, beliefs about infant sleep, and infant sleep duration; maternal depressive symptoms remained stable

Note: IBP = Infant Behavior Program; BB = Building Blocks; VIP = Video Interaction Project; ROR = Reach Out and Read; PIP = Psychoanalytic Parent-Infant Psychotherapy; FES-BSBW = Better Sleep Better Well-Being program; CHW = Community Health Worker program; ECD = Early Childhood Development program.

Declaration

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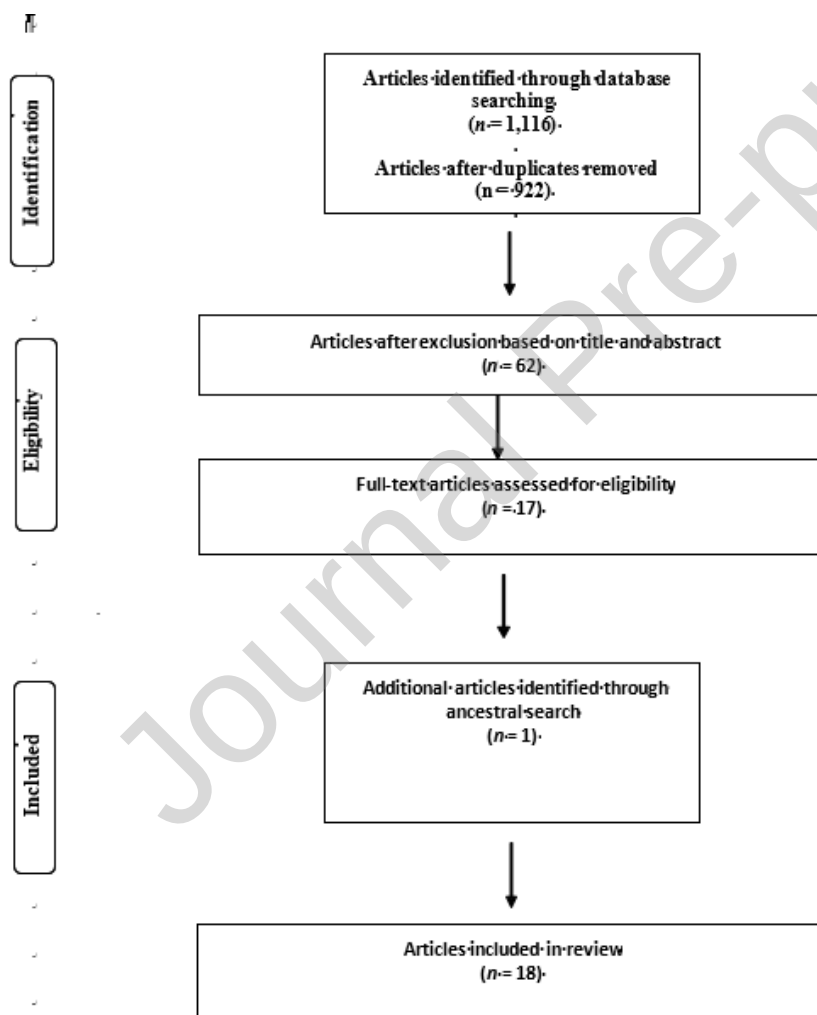


Figure 1: Search and article selection flow diagram. Adapted from Moher, Liberati, Tetzlaff, & Altman (2009)